KUZLAB - KUZFOLLOW - GITHUB

KUZFOLLOW - Complete Installation Guide (Debian, Apache2, Certbot, no PHP-FPM)

Assumptions

- OS: **fresh Debian server**
- Domain: **kuzfollow.kuzlab.org** (DNS already points to this server)
- Web stack: **Apache2 + mod_php** (no PHP-FPM)
- Deploy path: **/var/www/html/KuzFollow**
- Auth: **HTTP Basic Auth**
- Persistence: **local JSON** snapshots in `/var/www/html/KuzFollow/data`
- Commands below are executed **as root** (no `sudo`).

---

1) Base system preparation

1.1 Update and install required packages

```
apt update
apt -y upgrade

apt -y install \
  apache2 \
  php libapache2-mod-php php-curl \
  git curl ca-certificates \
  apache2-utils \
  certbot python3-certbot-apache
```

1.2 Enable Apache modules

```
a2enmod ssl rewrite headers
systemctl restart apache2
```

---

2) Deploy code (Git clone)

2.1 Clone repository into the Apache webroot

```
cd /var/www/html
git clone https://github.com/Kusanagi8200/KuzFollow.git KuzFollow
```

2.2 Create required directories

```
mkdir -p /var/www/html/KuzFollow/{public/assets,src,data,cache}
```

2.3 Permissions (write only where needed)

- Apache user `www-data` must write to `data/` and `cache/`.
- Keep everything else under root ownership.

```
chown -R root:root /var/www/html/KuzFollow
chown -R www-data:www-data /var/www/html/KuzFollow/{data,cache,public,src}
chmod 750 /var/www/html/KuzFollow/data /var/www/html/KuzFollow/cache
```

---

3) Secrets (outside the webroot)

3.1 Create config directory

```
mkdir -p /etc/kuzfollow
```

3.2 Create `/etc/kuzfollow/config.php`

> Put your real GitHub token here (recommend a fine-grained token with minimal scopes).

```
cat > /etc/kuzfollow/config.php <<'PHP'
<?php
return [
  'github_user'  => 'Kusanagi8200',
  'github_token' => 'CHANGE_ME_GITHUB_TOKEN',
];
PHP
```

3.3 Restrict permissions (Apache can read, not world-readable)

```
chown root:www-data /etc/kuzfollow/config.php
chmod 640 /etc/kuzfollow/config.php
```

---

4) HTTP Basic Auth

4.1 Create htpasswd file

```
htpasswd -c /etc/apache2/.htpasswd-kuzfollow admin
```

4.2 Secure htpasswd file

```
chown root:www-data /etc/apache2/.htpasswd-kuzfollow
chmod 640 /etc/apache2/.htpasswd-kuzfollow
```

---

5) Create application files (PHP + assets)

> These steps create a minimal PHP app:

- Dashboard (followers/following/repositories/events)
- Follow/unfollow actions with **dry-run**
- Daily follower snapshot (UTC) stored in JSON
- SVG graph generated from history
- KUZAI-style UI (neon + semi-transparent panels)
- Static full-site background image

5.1 `/var/www/html/KuzFollow/src/GitHubClient.php`

```
cat > /var/www/html/KuzFollow/src/GitHubClient.php <<'PHP'
<?php
declare(strict_types=1);

final class GitHubClient {
  private string $token;
  private string $api = 'https://api.github.com';

  public function __construct(string $token) { $this->token = $token; }

  private function request(string $method, string $path, ?array $body = null): array {
    $ch = curl_init($this->api.$path);
```

```
    $headers = [
      'Accept: application/vnd.github+json',
      'User-Agent: kuzfollow',
      'Authorization: Bearer '.$this->token,
    ];

    curl_setopt_array($ch, [
      CURLOPT_RETURNTRANSFER => true,
      CURLOPT_CUSTOMREQUEST  => $method,
      CURLOPT_HTTPHEADER     => $headers,
      CURLOPT_TIMEOUT        => 25,
    ]);

    if ($body !== null) {
      $payload = json_encode($body, JSON_UNESCAPED_SLASHES);
      curl_setopt($ch, CURLOPT_POSTFIELDS, $payload);
      curl_setopt($ch, CURLOPT_HTTPHEADER, array_merge($headers, ['Content-Type:
application/json']));
    }

    $resp = curl_exec($ch);
    $code = (int)curl_getinfo($ch, CURLINFO_RESPONSE_CODE);
    curl_close($ch);

    $json = json_decode($resp ?: '[]', true);
    if ($code >= 400) {
      $msg = is_array($json) && isset($json['message']) ? $json['message'] : 'GitHub API
error';
      throw new RuntimeException($msg.' (HTTP '.$code.')');
    }

    return is_array($json) ? $json : [];
  }

  public function followers(string $user): array {
    return $this->request('GET', "/users/{$user}/followers?per_page=100");
  }

  public function following(string $user): array {
    return $this->request('GET', "/users/{$user}/following?per_page=100");
```

```
  }

  public function reposOwnerSorted(string $user): array {
    return $this->request('GET', "/users/{$user}/repos?per_page=100&sort=updated");
  }

  public function eventsPublic(string $user, int $perPage = 30): array {
    $perPage = max(1, min(100, $perPage));
    return $this->request('GET', "/users/{$user}/events/public?per_page={$perPage}");
  }

  public function follow(string $target): void {
    $this->request('PUT', "/user/following/{$target}");
  }

  public function unfollow(string $target): void {
    $this->request('DELETE', "/user/following/{$target}");
  }
}
PHP
```

5.2 `/var/www/html/KuzFollow/public/.htaccess`

```
cat > /var/www/html/KuzFollow/public/.htaccess <<'HT'
RewriteEngine On

RewriteCond %{REQUEST_FILENAME} -f [OR]
RewriteCond %{REQUEST_FILENAME} -d
RewriteRule ^ - [L]

RewriteRule ^ index.php [QSA,L]
HT
```

5.3 `/var/www/html/KuzFollow/public/graph.svg.php`

```
cat > /var/www/html/KuzFollow/public/graph.svg.php <<'PHP'
<?php
declare(strict_types=1);

header('Content-Type: image/svg+xml; charset=utf-8');
header('Cache-Control: no-store');
```

```
function esc(string $s): string { return htmlspecialchars($s, ENT_QUOTES, 'UTF-8'); }

$W = 980; $H = 300;
$padL = 76; $padR = 22; $padT = 86; $padB = 58;
$plotW = $W - $padL - $padR;
$plotH = $H - $padT - $padB;

$pathA = __DIR__ . '/../data/followers_history.json';
$pathB = __DIR__ . '/followers_history.json';
$historyPath = is_file($pathA) ? $pathA : (is_file($pathB) ? $pathB : $pathA);

$history = [];
$err = null;

try {
  if (!is_file($historyPath)) throw new RuntimeException('history file not found');
  if (!is_readable($historyPath)) throw new RuntimeException('history file not readable');

  $raw = file_get_contents($historyPath);
  if ($raw === false || trim($raw) === '') throw new RuntimeException('history file empty');

  $j = json_decode($raw, true);
  if (!is_array($j)) throw new RuntimeException('history json invalid');

  foreach ($j as $p) {
    if (!is_array($p)) continue;
    $d = (string)($p['date'] ?? '');
    $v = (int)($p['followers'] ?? -1);
    if ($d === '' || $v < 0) continue;
    $history[] = ['date' => $d, 'followers' => $v];
  }

  if (count($history) < 2) throw new RuntimeException('not enough points (<2)');
} catch (Throwable $e) {
  $err = $e->getMessage();
}

echo '<?xml version="1.0" encoding="UTF-8"?>';
```

```
if ($err !== null) {
  $size = is_file($historyPath) ? (string)filesize($historyPath) : 'N/A';
  ?>
<svg xmlns="http://www.w3.org/2000/svg" width="<?= $W ?>" height="<?= $H ?>" viewBox="0 0 <?=
$W ?> <?= $H ?>">
  <rect x="0" y="0" width="<?= $W ?>" height="<?= $H ?>" fill="rgba(120,200,255,0.12)"
stroke="rgba(120,220,255,0.55)"/>
  <text x="18" y="32" fill="#7fdcff" font-family="system-ui,Segoe UI,Roboto" font-size="13"
letter-spacing="2">FOLLOWERS OVER TIME</text>
  <text x="18" y="62" fill="#ffffff" font-family="ui-monospace,Menlo,Consolas" font-
size="12"><?= esc("ERROR: ".$err) ?></text>
  <text x="18" y="86" fill="#cfefff" font-family="ui-monospace,Menlo,Consolas" font-
size="11"><?= esc("USING: ".$historyPath." (".$size." bytes)") ?></text>
</svg>
<?php
  exit;
}

$vals = array_map(fn($p) => (int)$p['followers'], $history);
$minV = min($vals); $maxV = max($vals);
if ($minV === $maxV) { $minV -= 1; $maxV += 1; }

$n = count($history);
$pts = [];
for ($i=0; $i<$n; $i++) {
  $x = $padL + ($plotW * ($i / max($n-1,1)));
  $v = (int)$history[$i]['followers'];
  $t = ($v - $minV) / ($maxV - $minV);
  $y = $padT + ($plotH * (1.0 - $t));
  $pts[] = [$x, $y, $v, (string)$history[$i]['date']];
}

$poly = implode(' ', array_map(fn($p)=>round($p[0],2).','.round($p[1],2), $pts));
$first = $pts[0][2];
$last  = $pts[$n-1][2];
$delta = $last - $first;
$sign  = $delta >= 0 ? '+' : '';
$rangeLabel = $pts[0][3] . ' → ' . $pts[$n-1][3];
$nowLabel = "now {$last} ({$sign}{$delta})";
$todayUTC = gmdate('Y-m-d');
```

```
$pill = function(float $x, float $y, string $text, int $fs = 12, string $fill = '#ffffff'):
string {
  $padX = 10; $padY = 7;
  $w = max(60, (int)(strlen($text) * ($fs * 0.62)) + $padX*2);
  $h = $fs + $padY*2;
  $rx = 10;
  $tX = $x + $padX;
  $tY = $y + $fs + $padY - 2;

  return
    '<rect x="'.round($x,2).'" y="'.round($y,2).'" width="'.$w.'" height="'.$h.'" rx="'.$rx.'"
fill="rgba(0,0,0,0.40)" stroke="rgba(120,220,255,0.55)"/>'.
    '<text x="'.round($tX,2).'" y="'.round($tY,2).'" fill="'.$fill.'" font-family="ui-
monospace,Menlo,Consolas" font-size="'.$fs.'">'.esc($text).'</text>';
};

$areaD = 'M '.$padL.' '.($padT+$plotH).' L ';
foreach ($pts as $p) $areaD .= round($p[0],2).' '.round($p[1],2).' L ';
$areaD .= ($padL+$plotW).' '.($padT+$plotH).' Z';

$lp = $pts[$n-1];
?>
<svg xmlns="http://www.w3.org/2000/svg" width="<?= $W ?>" height="<?= $H ?>" viewBox="0 0 <?=
$W ?> <?= $H ?>">
  <defs>
    <linearGradient id="area" x1="0" y1="0" x2="0" y2="1">
      <stop offset="0" stop-color="rgba(120,220,255,0.32)"/>
      <stop offset="1" stop-color="rgba(120,220,255,0.05)"/>
    </linearGradient>
    <filter id="glow" x="-40%" y="-40%" width="180%" height="180%">
      <feGaussianBlur stdDeviation="2.6" result="b"/>
      <feMerge><feMergeNode in="b"/><feMergeNode in="SourceGraphic"/></feMerge>
    </filter>
    <clipPath id="clipPlot">
      <rect x="<?= $padL ?>" y="<?= $padT ?>" width="<?= $plotW ?>" height="<?= $plotH ?>"
rx="10"/>
    </clipPath>
  </defs>
```

```
  <rect x="0" y="0" width="<?= $W ?>" height="<?= $H ?>" fill="rgba(120,200,255,0.12)" stroke="rgba(120,220,255,0.55)"/>

  <text x="18" y="30" fill="#7fdcff" font-family="system-ui,Segoe UI,Roboto" font-size="13" letter-spacing="2">FOLLOWERS OVER TIME</text>
  <?= $pill(18, 40, $rangeLabel, 12, '#ffffff') ?>
  <?= $pill($W-260, 40, $nowLabel, 12, '#ffffff') ?>

  <rect x="<?= $padL ?>" y="<?= $padT ?>" width="<?= $plotW ?>" height="<?= $plotH ?>" rx="12" fill="rgba(120,200,255,0.10)" stroke="rgba(255,255,255,0.10)"/>

  <?php for ($g=0; $g<=4; $g++):
    $y = $padT + ($plotH * ($g/4));
    $v = (int)round($maxV - (($maxV-$minV)*($g/4)));
  ?>
    <line x1="<?= $padL ?>" y1="<?= round($y,2) ?>" x2="<?= $padL+$plotW ?>" y2="<?= round($y,2) ?>" stroke="rgba(255,255,255,0.10)"/>
    <text x="<?= $padL-12 ?>" y="<?= round($y+4,2) ?>" text-anchor="end" fill="#cfefff" font-family="ui-monospace,Menlo,Consolas" font-size="11"><?= $v ?></text>
  <?php endfor; ?>

  <g clip-path="url(#clipPlot)">
    <path d="<?= esc($areaD) ?>" fill="url(#area)"/>
    <polyline points="<?= $poly ?>" fill="none" stroke="#33ccff" stroke-width="2.6" filter="url(#glow)"/>
  </g>

  <circle cx="<?= round($lp[0],2) ?>" cy="<?= round($lp[1],2) ?>" r="5.0" fill="#ffffff" stroke="#33ccff" stroke-width="2"/>

  <?= $pill(18, $H-44, "daily snapshot (UTC): {$todayUTC}", 11, '#cfefff') ?>
</svg>
PHP
```

5.4 `/var/www/html/KuzFollow/public/seed_history.php` (initial coherent curve)

- Creates **30 points** from **today-29 days → today**
- Growth: **+3 followers/day**
- Ends exactly at current `followers_now`

```
cat > /var/www/html/KuzFollow/public/seed_history.php <<'PHP'
<?php
declare(strict_types=1);

header('Content-Type: text/plain; charset=utf-8');

$config = require '/etc/kuzfollow/config.php';
require __DIR__ . '/../src/GitHubClient.php';

$user  = (string)($config['github_user'] ?? '');
$token = (string)($config['github_token'] ?? '');
if ($user === '' || $token === '') { http_response_code(500); exit("Missing config\n"); }

$gh = new GitHubClient($token);
$followers = $gh->followers($user);
$N = count($followers);

$points = 30;
$incPerDay = 3;

$tz = new DateTimeZone('UTC');
$today = new DateTimeImmutable('today', $tz);
$start = $today->sub(new DateInterval('P'.($points-1).'D'));

$startValue = $N - (($points-1) * $incPerDay);
if ($startValue < 0) $startValue = 0;

$hist = [];
for ($i=0; $i<$points; $i++) {
  $d = $start->add(new DateInterval('P'.$i.'D'))->format('Y-m-d');
  $v = $startValue + ($i * $incPerDay);
  if ($v > $N) $v = $N;
  $hist[] = ['date' => $d, 'followers' => $v];
}

$hist[$points-1]['date'] = $today->format('Y-m-d');
$hist[$points-1]['followers'] = $N;

@mkdir(__DIR__ . '/../data', 0755, true);
$file = __DIR__ . '/../data/followers_history.json';
```

```
file_put_contents($file, json_encode($hist, JSON_UNESCAPED_SLASHES));

echo "OK\n";
echo "user={$user}\n";
echo "points=".count($hist)."\n";
echo "start=".$hist[0]['date']."\n";
echo "end=".$hist[count($hist)-1]['date']."\n";
echo "followers_now={$N}\n";
echo "file={$file}\n";
PHP
```

5.5 `/var/www/html/KuzFollow/public/index.php`

```
cat > /var/www/html/KuzFollow/public/index.php <<'PHP'
<?php
declare(strict_types=1);

session_start();
require __DIR__ . '/../src/GitHubClient.php';
$config = require '/etc/kuzfollow/config.php';

function h(string $s): string { return htmlspecialchars($s, ENT_QUOTES, 'UTF-8'); }

$user  = (string)($config['github_user'] ?? '');
$token = (string)($config['github_token'] ?? '');
if ($user === '' || $token === '') { http_response_code(500); exit('Server not configured'); }

$gh = new GitHubClient($token);
$preview = null;

/* actions + dry-run */
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
  $action = (string)($_POST['action'] ?? '');
  $users  = $_POST['users'] ?? [];
  $dry    = isset($_POST['dry']);

  if (!is_array($users)) $users = [];
  $users = array_values(array_unique(array_values(array_filter(array_map('trim',
array_map('strval', $users))))));
```

```
  if ($dry) {
    $preview = ['action' => $action, 'users' => $users];
  } else {
    foreach ($users as $u) {
      if ($action === 'follow')   $gh->follow($u);
      if ($action === 'unfollow') $gh->unfollow($u);
      usleep(300000);
    }
    header('Location: /');
    exit;
  }
}

/* data */
$followers = $gh->followers($user);
$following = $gh->following($user);
$repos     = $gh->reposOwnerSorted($user);
$events    = $gh->eventsPublic($user, 30);

$fol = array_column($followers, 'login');
$ing = array_column($following, 'login');

$youDontFollowBack = array_values(array_diff($fol, $ing)); // followers you don't follow
$theyDontFollowYou = array_values(array_diff($ing, $fol)); // following who don't follow back

sort($youDontFollowBack);
sort($theyDontFollowYou);

$recentFollowers = array_slice($followers, 0, 10);

/* daily snapshot (UTC) written on first visit of the day */
$dataDir = __DIR__ . '/../data';
$historyFile = $dataDir . '/followers_history.json';
@mkdir($dataDir, 0755, true);

$today = gmdate('Y-m-d');
$countFollowers = count($followers);

$hist = [];
if (is_file($historyFile)) {
```

```
  $raw = file_get_contents($historyFile);
  $j = json_decode($raw ?: '[]', true);
  if (is_array($j)) $hist = $j;
}
$last = end($hist);
$needsAppend = !is_array($last) || (($last['date'] ?? '') !== $today);
if ($needsAppend) {
  $hist[] = ['date' => $today, 'followers' => $countFollowers];
  if (count($hist) > 180) $hist = array_slice($hist, -180);
  file_put_contents($historyFile, json_encode(array_values($hist), JSON_UNESCAPED_SLASHES));
}
?>
<!doctype html>
<html>
<head>
<meta charset="utf-8">
<title>KUZFOLLOW</title>
<meta name="viewport" content="width=device-width, initial-scale=1">
<link rel="stylesheet" href="/assets/style.css">
</head>
<body>
<div class="wrap">

<div class="card" style="text-align:center;">
  <div style="font-size:22px;font-weight:700;letter-spacing:0.18em;color:#ffffff;text-
transform:uppercase;text-shadow:0 0 6px rgba(51,204,255,.6),0 0 14px rgba(51,204,255,.4),0 0
28px rgba(51,204,255,.2);">
    KUZFOLLOW GITHUB
  </div>
  <div style="margin-top:6px;font-size:13px;letter-spacing:0.22em;color:#cfefff;opacity:.95;">
    A KUZ NETWORK SOLUTION — [BETA-0.1.2025]
  </div>
</div>

<div class="card">
  <h2>KUZFOLLOW · <?=h($user)?></h2>
  <div class="kv">
    <span class="pill">followers <?=count($followers)?></span>
    <span class="pill">following <?=count($following)?></span>
    <span class="pill">repos <?=count($repos)?></span>
```

```
    <span class="pill"><a href="#" onclick="openEvents();return false;">events</a></span>
  </div>
</div>

<?php if ($preview): ?>
<div class="card">
  <h2>DRY-RUN PREVIEW</h2>
  <div class="meta">action: <?=h($preview['action'])?> · targets:
<?=count($preview['users'])?></div>
  <?php foreach ($preview['users'] as $u): ?>
    <div class="item"><?=h($u)?></div>
  <?php endforeach; ?>
</div>
<?php endif; ?>

<div class="grid">

  <div class="card">
    <h2>10 RECENT FOLLOWERS</h2>
    <?php if (!$recentFollowers): ?>
      <div class="item"><div class="meta">none</div></div>
    <?php else: ?>
      <?php foreach ($recentFollowers as $f): ?>
        <div class="item"><?=h((string)($f['login'] ?? ''))?></div>
      <?php endforeach; ?>
    <?php endif; ?>
  </div>

  <div class="card">
    <h2>REPOSITORIES (OWNER / UPDATED)</h2>
    <?php foreach (array_slice($repos, 0, 12) as $r): ?>
      <div class="item">
        <a href="<?=h((string)($r['html_url'] ?? '#'))?>"
target="_blank"><?=h((string)($r['name'] ?? 'repo'))?></a>
        <div class="meta">★<?= (int)($r['stargazers_count'] ?? 0) ?> · forks <?=
(int)($r['forks_count'] ?? 0) ?> · <?=h((string)($r['language'] ?? ''))?></div>
      </div>
    <?php endforeach; ?>
  </div>
```

```
  <div class="card">
    <h2>FOLLOW BACK</h2>
    <form method="post">
      <input type="hidden" name="action" value="follow">
      <?php if (!$youDontFollowBack): ?>
        <div class="item"><div class="meta">none</div></div>
      <?php else: ?>
        <?php foreach ($youDontFollowBack as $u): ?>
          <div class="item"><label><input type="checkbox" name="users[]" value="<?=h($u)?>">
<?=h($u)?></label></div>
        <?php endforeach; ?>
      <?php endif; ?>
      <label class="meta"><input type="checkbox" name="dry"> dry-run</label><br>
      <button class="btn">FOLLOW SELECTED</button>
    </form>
  </div>

  <div class="card">
    <h2>UNFOLLOW (NON MUTUAL)</h2>
    <form method="post">
      <input type="hidden" name="action" value="unfollow">
      <?php if (!$theyDontFollowYou): ?>
        <div class="item"><div class="meta">none</div></div>
      <?php else: ?>
        <?php foreach ($theyDontFollowYou as $u): ?>
          <div class="item"><label><input type="checkbox" name="users[]" value="<?=h($u)?>">
<?=h($u)?></label></div>
        <?php endforeach; ?>
      <?php endif; ?>
      <label class="meta"><input type="checkbox" name="dry"> dry-run</label><br>
      <button class="btn alt">UNFOLLOW SELECTED</button>
    </form>
  </div>

  <div class="card" style="grid-column:1/-1;">
    <h2>FOLLOWERS OVER TIME</h2>
    <div class="item" style="padding:10px;">
      <img src="/graph.svg.php" alt="Followers over time"
style="width:100%;height:auto;display:block;">
    </div>
```

```
  </div>

</div>
</div>

<div id="events" class="modal">
  <div class="box">
    <div class="head">
      <h2>EVENTS</h2>
      <button class="btn" onclick="closeEvents()">CLOSE</button>
    </div>
    <div class="body">
      <?php foreach (array_slice($events, 0, 30) as $e): ?>
        <div class="item">
          <?=h((string)($e['type'] ?? 'Event'))?>
          <div class="meta"><?=h((string)($e['repo']['name'] ?? ''))?></div>
        </div>
      <?php endforeach; ?>
    </div>
  </div>
</div>

<script>
function openEvents(){document.getElementById('events').classList.add('open')}
function closeEvents(){document.getElementById('events').classList.remove('open')}
</script>

</body>
</html>
PHP
```

5.6 `/var/www/html/KuzFollow/public/assets/style.css` (full CSS)

```
cat > /var/www/html/KuzFollow/public/assets/style.css <<'CSS'
:root{
  --neon:#33ccff;
  --neon2:#7fdcff;
  --txt:#ffffff;
  --muted:#cfefff;
```

```
  --panel:rgba(120,200,255,0.18);
  --panel2:rgba(120,200,255,0.12);
  --line:rgba(120,220,255,0.55);

  --radius:16px;
}

*{ box-sizing:border-box; }

html{ min-height:100%; }

body{
  min-height:100%;
  margin:0;
  color:var(--txt);
  font-family:system-ui, Segoe UI, Roboto;

  /* Static full-site background */
  background-image:
    linear-gradient(180deg, rgba(5,20,40,0.85), rgba(5,20,40,0.95)),
    radial-gradient(circle at top, rgba(51,204,255,0.18), transparent 60%),
    url("/assets/bg.jpg");

  background-position:center center;
  background-repeat:no-repeat;
  background-size:cover;
  background-attachment:fixed;
  background-blend-mode:normal, screen, normal;
  background-color:#050f1f;
}

.wrap{
  max-width:1200px;
  margin:26px auto;
  padding:0 14px;
}

.grid{
  display:grid;
  grid-template-columns:1fr 1fr;
```

```
  gap:14px;
}

@media(max-width:900px){
  .grid{ grid-template-columns:1fr; }
}

.card{
  background:var(--panel);
  border:1px solid var(--line);
  border-radius:var(--radius);
  padding:14px;
  margin-bottom:14px;
  box-shadow:0 0 18px rgba(51,204,255,.25);
  backdrop-filter:blur(10px);
}

.card h2{
  margin:0 0 10px;
  font-size:13px;
  font-weight:600;
  letter-spacing:0.14em;
  text-transform:uppercase;
  color:var(--neon2);
}

.item{
  background:var(--panel2);
  border:1px solid rgba(255,255,255,0.25);
  border-radius:12px;
  padding:8px 10px;
  margin-bottom:6px;
}

.meta{
  font-size:12px;
  color:var(--muted);
}

a{ color:var(--neon); text-decoration:none; }
```

```
a:hover{ text-decoration:underline; }

.pill{
  display:inline-block;
  border:1px solid var(--line);
  border-radius:999px;
  padding:4px 10px;
  margin-right:6px;
  font-size:12px;
  color:var(--txt);
  background:rgba(0,0,0,0.35);
}

.btn{
  margin-top:8px;
  padding:8px 12px;
  border-radius:12px;
  border:1px solid var(--line);
  background:rgba(0,0,0,0.35);
  color:#ffffff;
  cursor:pointer;
}

.btn:hover{ background:rgba(51,204,255,0.25); }
.btn.alt{ background:rgba(120,200,255,0.25); }

.kv{ font-size:12px; color:var(--muted); }

.modal{
  position:fixed;
  inset:0;
  display:none;
  background:rgba(0,0,0,0.65);
  z-index:10000;
}

.modal.open{
  display:flex;
  align-items:center;
  justify-content:center;
```

```
}

.modal .box{
  width:min(820px,92vw);
  max-height:80vh;
  background:var(--panel);
  border:1px solid var(--line);
  border-radius:18px;
  padding:14px;
  overflow:hidden;
  backdrop-filter:blur(10px);
}

.modal .head{
  display:flex;
  justify-content:space-between;
  align-items:center;
}

.modal .body{
  margin-top:10px;
  max-height:60vh;
  overflow:auto;
}

img{ max-width:100%; height:auto; display:block; }
CSS
```

5.7 Background image

Place your background file here:

- `/var/www/html/KuzFollow/public/assets/bg.jpg`

---

6) Apache VirtualHost + HTTPS (Certbot)

6.1 HTTP vhost (port 80)

Create `/etc/apache2/sites-available/kuzfollow.kuzlab.org.conf`:

```
cat > /etc/apache2/sites-available/kuzfollow.kuzlab.org.conf <<'AP'
<VirtualHost *:80>
  ServerName kuzfollow.kuzlab.org
  DocumentRoot /var/www/html/KuzFollow/public

  <Directory /var/www/html/KuzFollow/public>
    Options FollowSymLinks
    AllowOverride All
    Require all granted
  </Directory>

  ErrorLog ${APACHE_LOG_DIR}/kuzfollow-error.log
  CustomLog ${APACHE_LOG_DIR}/kuzfollow-access.log combined
</VirtualHost>
AP
```

Enable and reload:

```
a2ensite kuzfollow.kuzlab.org.conf
systemctl reload apache2
```

6.2 Issue TLS cert via Certbot (Apache plugin)

```
certbot --apache -d kuzfollow.kuzlab.org
```

6.3 Force Basic Auth on HTTPS vhost

After Certbot, update `/etc/apache2/sites-available/kuzfollow.kuzlab.org-le-ssl.conf` to:

```
cat > /etc/apache2/sites-available/kuzfollow.kuzlab.org-le-ssl.conf <<'AP'
<IfModule mod_ssl.c>
<VirtualHost *:443>
  ServerName kuzfollow.kuzlab.org
  DocumentRoot /var/www/html/KuzFollow/public

  <Directory /var/www/html/KuzFollow/public>
    Options FollowSymLinks
    AllowOverride All
    Require all granted

    AuthType Basic
```

```
    AuthName "KuzFollow Restricted"
    AuthUserFile /etc/apache2/.htpasswd-kuzfollow
    Require valid-user
  </Directory>

  ErrorLog ${APACHE_LOG_DIR}/kuzfollow-error.log
  CustomLog ${APACHE_LOG_DIR}/kuzfollow-access.log combined

  SSLEngine on
  SSLCertificateFile /etc/letsencrypt/live/kuzfollow.kuzlab.org/fullchain.pem
  SSLCertificateKeyFile /etc/letsencrypt/live/kuzfollow.kuzlab.org/privkey.pem
  Include /etc/letsencrypt/options-ssl-apache.conf
</VirtualHost>
</IfModule>
AP
```

Enable and reload:

```
a2ensite kuzfollow.kuzlab.org-le-ssl.conf
systemctl reload apache2
```

---

7) Initialize history (seed) and validate

7.1 Seed a coherent initial curve

```
curl -k -u 'admin:YOUR_PASSWORD' https://kuzfollow.kuzlab.org/seed_history.php
```

7.2 Validate SVG graph

```
curl -k -u 'admin:YOUR_PASSWORD' https://kuzfollow.kuzlab.org/graph.svg.php | head -n 20
```

7.3 Check the history file exists

```
ls -l /var/www/html/KuzFollow/data/followers_history.json
head -n 3 /var/www/html/KuzFollow/data/followers_history.json
```

---

8) How the daily update works (real data)

- Each time someone opens `/` (index.php), the app:
  - fetches the current follower count from GitHub
  - appends **one snapshot per day (UTC)** if today’s date is not already present

- If no one visits the page on a given day, no snapshot is written (optional cron below).

---

9) Certbot renew verification

```
systemctl list-timers | grep certbot || true
certbot renew --dry-run
```

---

10) Optional: guaranteed daily snapshot (even with no visits)

Create a cron job that triggers the page once per day (UTC):

```
cat > /etc/cron.d/kuzfollow-snapshot <<'CR'
### Snapshot daily at 00:10 UTC
10 0 * * * root curl -k -u 'admin:YOUR_PASSWORD' https://kuzfollow.kuzlab.org/ > /dev/null
2>&1
CR
```

---

11) Quick troubleshooting

11.1 vhost routing

```
apachectl -S
```

11.2 Logs

```
tail -n 200 /var/log/apache2/kuzfollow-error.log
tail -n 200 /var/log/apache2/kuzfollow-access.log
```

11.3 Enabled modules

```
apachectl -M | egrep 'ssl|rewrite|headers|php'
```

File tree recap

- `/var/www/html/KuzFollow/src/GitHubClient.php`
- `/var/www/html/KuzFollow/public/index.php`
- `/var/www/html/KuzFollow/public/graph.svg.php`
- `/var/www/html/KuzFollow/public/seed_history.php`
- `/var/www/html/KuzFollow/public/.htaccess`
- `/var/www/html/KuzFollow/public/assets/style.css`
- `/var/www/html/KuzFollow/public/assets/bg.jpg`
- `/var/www/html/KuzFollow/data/followers_history.json` (created by seed / daily snapshot)

- `/etc/kuzfollow/config.php` (secrets)
- `/etc/apache2/.htpasswd-kuzfollow` (basic auth)

Revision #3
Created 14 December 2025 17:55:34 by Kusanagi8200
Updated 2 January 2026 15:54:19 by Kusanagi8200